

A bucket-brigade quantum random access memory

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Quantum random access memory (QRAM) enables efficient access to classical data for quantum computers and is a prerequisite for many quantum algorithms in achieving quantum speed-up. Despite various proposals, there have not been many experimental realizations of QRAM. Here we use a superconducting quantum processor to implement a circuit-based bucket-brigade QRAM, which uses a binary tree of quantum routers to enable efficient addressing of the stored information. To facilitate the experimental implementation, we introduce an efficient gate decomposition scheme for quantum routers, which effectively reduces the depth of the QRAM circuit compared with the conventional controlled-SWAP implementation. We further propose an error mitigation method to improve the query fidelity of the QRAM. With these techniques, we are able to experimentally implement the QRAM architectures for addressing four and eight classical bits, achieving query fidelities up to 0.809 ± 0.025 and 0.604 ± 0.005 , respectively. Additionally, we study the error propagation mechanism and the scalability of our QRAM implementation, which provides experimental evidence for the noise resilience of the bucket-brigade architecture. Our results highlight the potential of superconducting quantum processors for realizing a scalable QRAM architecture.

The ability to perform quantum queries of classical data is a fundamental requirement for numerous quantum algorithms, such as quantum machine learning^{1–6}, quantum simulation^{7–11}, Grover's algorithm¹² and a host of other areas^{13–16}. In practice, this querying process can often dominate the computational overhead, which notably impacts the potential quantum advantage¹⁷. Quantum random access memory (QRAM)^{18–30} offers a promising solution to this dilemma, as it allows quantum processors to access and retrieve data in superposition via quantum addresses. Considering a memory of size N where each

classical data value x_i is stored at address $i \in \{0, 1, \dots, N-1\}$, the QRAM performs the following unitary transformation:

$$\sum_{i=0}^{N-1} \alpha_i |i\rangle_A |0\rangle_D \xrightarrow{\text{QRAM}} \sum_{i=0}^{N-1} \alpha_i |i\rangle_A |x_i\rangle_D, \quad (1)$$

where α_i is the amplitude of each address in the superposition, and $| \cdot \rangle_A (| \cdot \rangle_D)$ is the address (data) qubit register storing the input (output). Notably, the above operation can be realized with a time complexity

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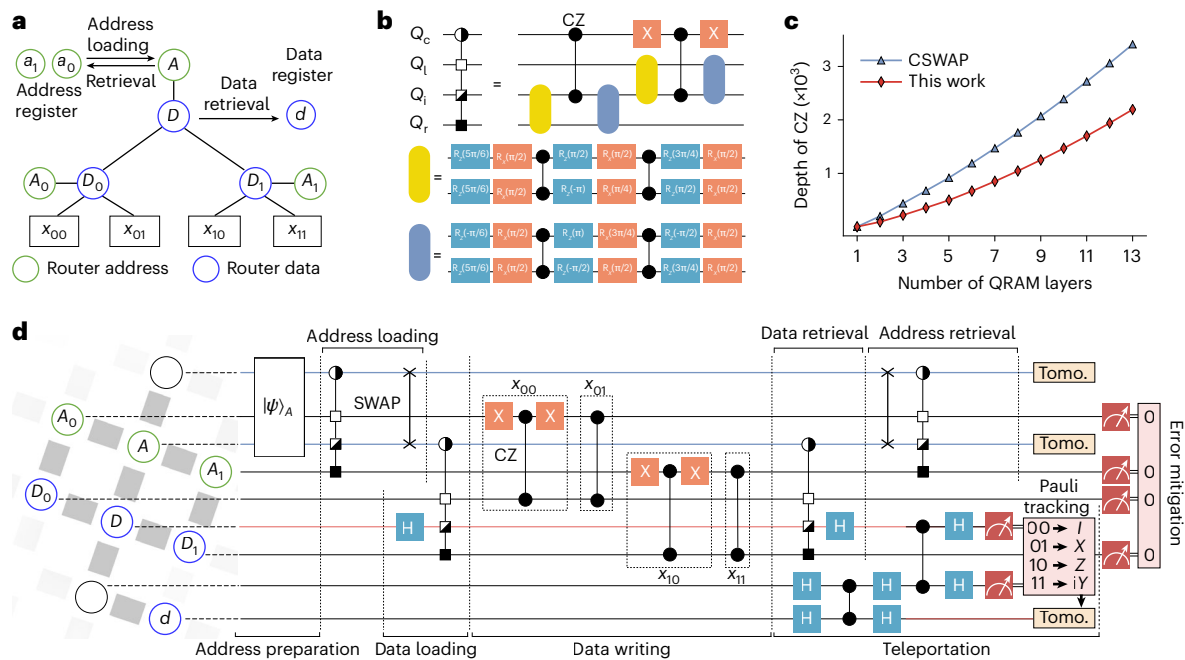


Fig. 1 | Experimental implementation of QRAM with a superconducting quantum processor. **a**, Schematic of the two-layer bucket-brigade QRAM, which can query four classical data bits, denoted as x_{00}, x_{01}, x_{10} and x_{11} , according to the query address. The qubits in the QRAM routers can be divided into two groups, one for storing the address information (router address, labelled by A), and the other for loading the data bus (router data, labelled by D). The data bus encodes the data bit in the $|\pm\rangle$ basis to ensure that the data-writing operation acts trivially on all qubits that do not contain the data bus, specifically, those remaining in the $|0\rangle$ state. **b**, The experimental circuit for implementing the quantum routing operation. The incident qubit (Q_i) is connected to the control qubit (Q_c) and two output (Q_i and Q_o) qubits simultaneously. The quantum routing operation swaps the quantum state of Q_i to Q_o when Q_c is in $|0\rangle$ and to Q_i when Q_c is in $|1\rangle$. Here $R_X(\theta)$ ($R_Z(\theta)$) represents the single-qubit rotation gate, which rotates the qubit state

around the X (Z) axis in the Bloch sphere by angle θ . For simplicity, we further denote the $R_X(\pi)$ gate as X . **c**, Comparison of CZ gate depths in the QRAM circuit for various numbers of layers, evaluated for the conventional CSWAP approach and our proposed hardware-efficient decomposition scheme. Here we consider the implementation based on the H-tree recursive mapping and quantum teleportation techniques. **d**, Schematic of the device layout and the corresponding quantum circuit used to benchmark the two-layer bucket-brigade QRAM. Physical qubits are labelled according to their respective roles during the data-loading, data-writing and data-retrieval phases. The initial address state $|\psi\rangle_A$ is prepared on the two address register qubits. For a detailed explanation of the role of each physical qubit in each phase and the operations performed, see Extended Data Fig. 1. Here H denotes the Hadamard gate. Tomo., tomography.

of $\mathcal{O}(\log N)$, which is optimal for implementing any oracle-based quantum algorithm²⁷.

Among various QRAM architecture designs, the bucket-brigade QRAM architecture¹⁸ is a promising approach due to its logarithmic scaling of infidelity with the memory size²¹. The architecture consists of a binary tree of quantum routers, each capable of coherently directing the input signal to superpositions of output paths based on quantum addresses³¹. Such quantum routers essentially involve four-body interactions, making their physical implementation challenging. Despite various theoretical proposals^{24,30,32}, experimental progress was limited until recent breakthroughs demonstrated deterministic quantum routers on superconducting platforms^{33,34}. Nevertheless, a scalable implementation of the full bucket-brigade QRAM architecture, along with a comprehensive understanding of its error propagation under realistic noise conditions, remains largely unexplored.

Here we report on the experimental realization and investigation of a scalable bucket-brigade QRAM architecture on a high-performance programmable superconducting quantum processor. The qubits are arranged in a two-dimensional square grid. By exploring the inherent degree of freedom in the QRAM, we propose an efficient gate decomposition scheme of the quantum routing operation, which reduces the circuit depth by more than 30% compared with a traditional routing circuit based on a controlled-SWAP (CSWAP) gate²⁴. With this, we present an end-to-end demonstration of the bucket-brigade QRAM architecture consisting of a full cycle of address loading, data loading, data writing, data retrieval and address retrieval. We first focus on the two-layer QRAM implementation (Fig. 1a), where we use quantum teleportation^{23,35} to transfer the queried state into the data register. We

comprehensively investigate its performance with different quantum addresses and classical data configurations, observing the correlation between the query fidelity and the number of address components. By using router qubits as error indicators, we develop an efficient error mitigation technique to enhance the query fidelity. We then move on to a three-layer QRAM, for which we are able to experimentally validate the localized nature of error propagation in the QRAM through observing its performance with various errors on different branches. Furthermore, by capturing the entanglement structure among the memory components, we experimentally identify the origin of the noise resilience of the bucket-brigade QRAM architecture. We finish by providing a scalability analysis of our QRAM implementation based on gate error models.

Experimental implementation

The digital implementation of quantum routers, which involves decomposing each router into native-gate quantum circuits that respect device connectivity constraints, is highly resource-intensive. Conventionally, a quantum router can be constructed using two CSWAP gates²¹, with each CSWAP requiring eight or ten CZ gates depending on the connectivity between the control qubit and target qubits³⁶. During QRAM operations, routing is confined to specific subspaces, leading to more circuit optimization opportunities (see Supplementary Section II B for details). By identifying a family of functionally equivalent unitaries for the routing operation, we select the most efficient variant when decomposed into the native gate set of arbitrary single-qubit gates and CZ gates between nearest-neighbour qubits. Our optimized implementation, shown in Fig. 1b, demonstrates a particularly efficient

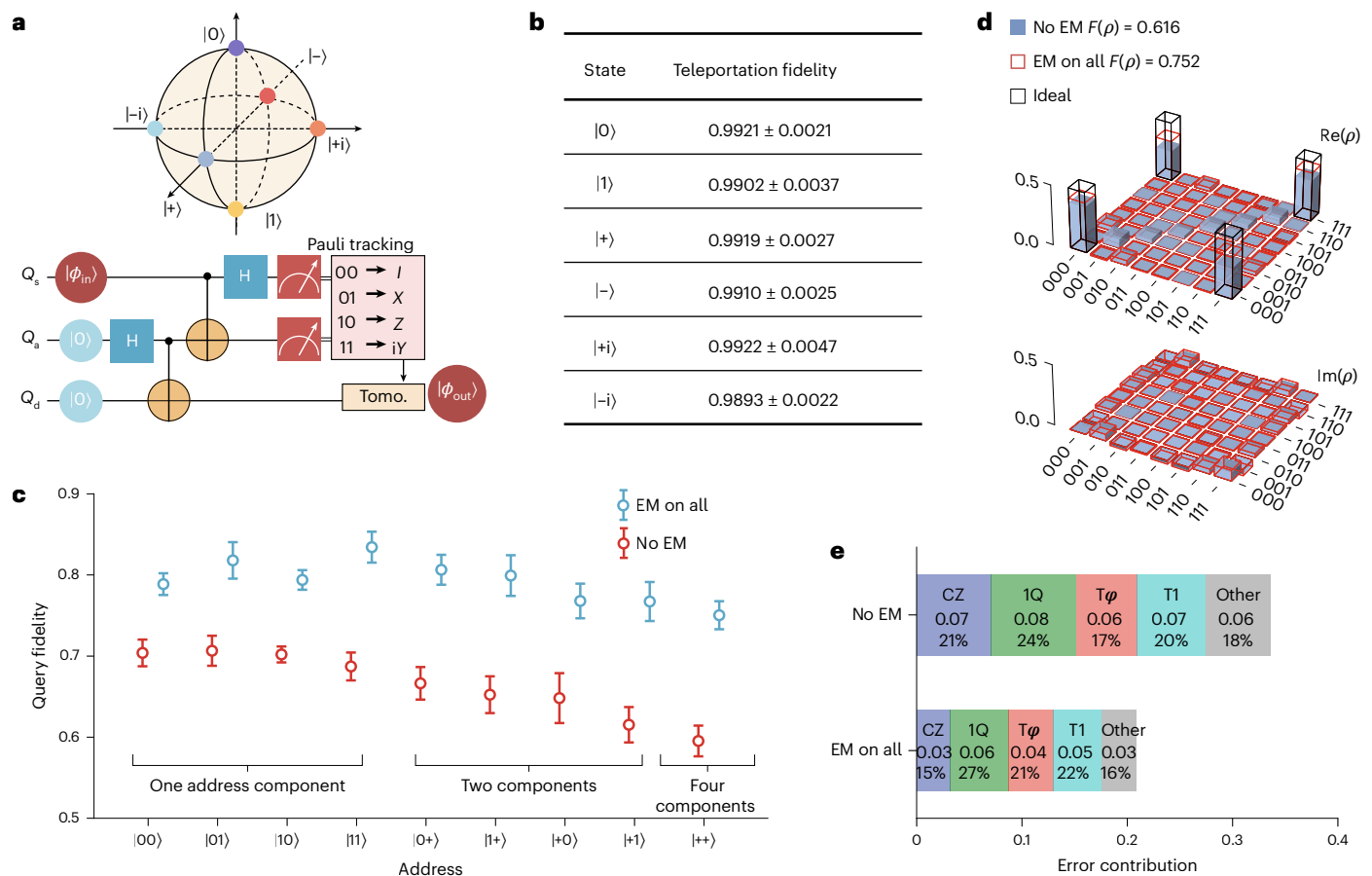


Fig. 2 | Experimental quantum teleportation and QRAM performances.

a, The six quantum states used in the teleportation experiment and the quantum circuit of teleportation. The input state $|\phi_{in}\rangle$ in the source qubit (Q_s) is transferred to the data qubit (Q_d) through the ancillary qubit (Q_a). The output state $|\phi_{out}\rangle$ is characterized using quantum state tomography. The tomography operation is reinterpreted based on the output of Q_s and Q_d via Pauli tracking. **b**, The teleportation fidelity values of the six quantum states, calculated as

$F \equiv (\text{Tr} \sqrt{\rho_{out}^{1/2} \sigma_{in} \rho_{out}^{1/2}})^2$, where ρ_{out} is the experimentally obtained output state and σ_{in} denotes the ideal input state. The uncertainty is given as the standard deviation from five repeated measurements. **c**, Query fidelity values of the two-layer QRAM for query addresses consisting of different numbers of

components, with and without error mitigation. The fidelity values are averaged over 16 classical data configurations, with the error bars being the corresponding standard deviation. **d**, Density matrices of GHZ states prepared using a two-layer bucket-brigade QRAM, with and without error mitigation techniques applied. The witness fidelity value for verifying genuine multipartite entanglement in GHZ state is 0.5. **e**, Error budget for the two-layer QRAM query infidelity. CZ is the CZ gate depolarizing error, excluding the decoherence error during CZ gates. 1Q is the single-qubit gate depolarizing error, excluding the decoherence error during single-qubit gates. T1 is the qubit energy relaxation error. T_ϕ is the qubit dephasing error. The results are obtained via numerical simulations, averaged over all addresses and classical data configurations. EM, error mitigation.

case where the incident qubit (Q_i) connects directly to the control (Q_c) and two output (Q_1 and Q_2) qubits. It requires only ten CZ gates, a 37.5% reduction compared with the CSWAP approach. The routing fidelity of this operation is characterized to be around 0.945 ± 0.009 , which is comparable with that reported in a recent study³³. For other connectivity configurations, the corresponding circuits are provided in Supplementary Fig. 6. Overall, our implementation achieves more than a 30% reduction in CZ gate depth, as illustrated in Fig. 1c. Although finding optimal decompositions of arbitrary unitaries has been proven to be an NP-complete problem³⁷, we believe that our decompositions are nearly optimal for implementing the routing operation on our device.

The complete quantum circuit for a two-layer bucket-brigade QRAM is depicted in Fig. 1d. It comprises five operational phases: address loading, data loading, data writing, data retrieval and address retrieval. A detailed explanation of the role of each physical qubit in each phase and the operations performed is shown in Extended Data Fig. 1. For the QRAM architecture on a square-grid device to exhibit the logarithmic time complexity illustrated in Fig. 1c, it will be necessary to integrate quantum teleportation in future large-scale implementations. Here, as a proof-of-principle demonstration, we employ

quantum teleportation at the end of the data-retrieval phase. After compilation, the circuit contains 41 layers of CZ gates interleaved with single-qubit gates. In our experiment, we realize median fidelities of about 99.96% and 99.7% for the parallel single- and two-qubit gates, which are characterized through simultaneous cross-entropy benchmarking^{38,39} (see Supplementary Section II A for more details on device performance). We also profile the performance of the quantum teleportation operation used in our experiment. Specifically, we prepare six distinct input states on the Bloch sphere, as shown in Fig. 2a, and measure the fidelities of the corresponding output states after quantum teleportation with Pauli tracking⁴⁰ and quantum state tomography (Methods). Figure 2b provides the results, which yield an average fidelity of 0.991 ± 0.003 .

QRAM performance

With the experimental bucket-brigade QRAM architecture established, we first investigate its performance under different configurations of the query address and classical data. We quantify the performance using the query fidelity metric, defined as $F = \langle \psi_{ideal} | \rho_{address,data} | \psi_{ideal} \rangle$, where $\rho_{address,data}$ is the density matrix of the data and address registers after the QRAM query and $|\psi_{ideal}\rangle$ is the corresponding ideal state.

For a two-layer QRAM, there are four classical data bits, each of which can be accessed by one of the query addresses $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ through the corresponding QRAM path. The classical data can also be queried in parallel by preparing the address qubits in an arbitrary quantum superposition of the four base components. In our experiment, we measure the query fidelities of the QRAM over all 16 different classical data configurations for different query addresses. The results are summarized in Fig. 2c. By grouping the query addresses according to the number of base components involved, we observe a clear degradation of the query fidelity with an increase in the number of components. Specifically, the query fidelity degrades from an average of 0.700 ± 0.018 for single-component addresses to 0.645 ± 0.031 and 0.595 ± 0.019 for two- and four-component addresses. The primary cause of this phenomenon is attributed to the dephasing error in our device (see Supplementary Section II E for a detailed analysis).

The router qubits in QRAM are disentangled from the address and data registers and remain in the ground state before and after a full cycle of query operations. As a result, we can measure the router qubits after the query without disturbing the quantum state in the address and data registers. Any populations detected in the router qubits signal the occurrence of an error. Based on this idea, we develop an error mitigation strategy to improve the query fidelity. By measuring the router qubits and discarding the events where any of the router qubits are detected in the excited state, we find a notable improvement in query fidelity among all query addresses, with the average values reaching 0.809 ± 0.025 , 0.785 ± 0.029 and 0.750 ± 0.017 for single-, two- and four-component addresses (Fig. 2c). See Extended Data Fig. 2 for the corresponding proportions of valid data. To visualize the effect of error mitigation on the output state, we prepare a Greenberger–Horne–Zeilinger (GHZ) state among the address and data qubits by setting the query address to $|\psi\rangle_A = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ and configuring the classical data bits as 0101. After applying error mitigation, we observe an increase in all four non-trivial terms in the density matrix (Fig. 2d), leading to a fidelity improvement of more than 0.13. In Fig. 2e, we present an error budget analysis of the query infidelity, comparing cases with and without error mitigation. The analysis accounts for energy relaxation, dephasing, single-qubit gate errors and CZ gate errors (see Supplementary Information for details about the error budget analysis). Our results indicate that the error mitigation strategy effectively reduces the influence of all error sources considered.

Notably, our approach provides a flexible tool for diagnosing the impact of error events that occur in different parts of the QRAM. By selectively applying error mitigation techniques to either the queried or unqueried branches of the QRAM architecture, we can evaluate their respective contributions to fidelity degradation. As demonstrated in Fig. 3 for the representative query addresses $|0+\rangle$ and $|1+\rangle$ (accessing the left-hand and right-hand halves of the QRAM, respectively), the fidelity improves by 0.113 when the branch is queried, compared with only 0.041 when it is not. A similar trend is observed for the right-hand branch, where the fidelity improvements are 0.104 and 0.057 for the queried and unqueried cases, respectively. This ‘shielding’ effect against errors on unqueried branches reflects the localized nature of error propagation in the bucket-brigade QRAM architecture. A comprehensive analysis of the data is provided in Supplementary Section II E.

Error propagation mechanism and query fidelity scaling

To gain a deeper understanding of the error propagation mechanism of the bucket-brigade QRAM, we investigate a three-layer implementation comprising 16 qubits (see Extended Data Fig. 3 for the experimental circuit). We focus on querying the leftmost two among the eight classical data bits of the QRAM, which are configured to 01010101 with the query address $|00+\rangle$. The query fidelity is measured to be

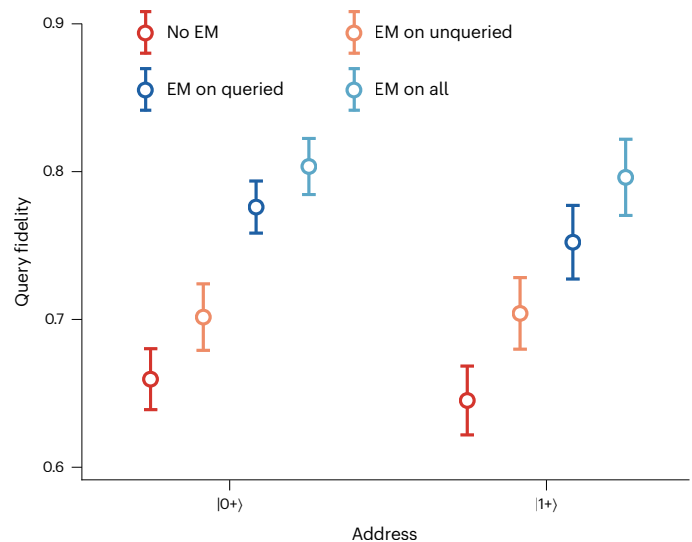


Fig. 3 | Query fidelity values for the query addresses $|0+\rangle$ and $|1+\rangle$, averaged over 16 different classical data configurations. Results are shown for the following cases: no error mitigation applied, error mitigation applied only to the queried branches, error mitigation applied only to the unqueried branches and error mitigation applied to all branches. Error bars represent the standard deviation of the query fidelity across the 16 classical data configurations.

0.604 ± 0.005 after applying error mitigation. The corresponding error budget is presented in Extended Data Fig. 4. By introducing depolarizing errors with tunable error rates to one of the four router qubits in the third layer, we quantitatively characterize the resulting degradation of the query fidelity. Experimentally, depolarizing errors are injected by randomly applying X, Y or Z gates to the target qubit, with a probability of $p/3$ each, after the data-loading phase (Extended Data Fig. 3). The depolarizing error rate is calculated as $e_d = 4p/3$.

The experimental results, presented in Fig. 4a, reveal distinct patterns of error susceptibility across different QRAM paths. When errors are introduced in the actively queried path, we observe a rapid, approximately linear degradation of query fidelity with increasing error rate (slope $k = -0.2549 \pm 0.0108$). By contrast, injecting errors into the three unqueried paths results in fidelity decay slopes of -0.0279 ± 0.0138 , -0.0301 ± 0.0082 and -0.0053 ± 0.0129 from left to right, respectively. This clear spatial dependence can be explained by the intrinsic error propagation mechanism of the bucket-brigade QRAM. Errors occurring in the leftmost branch can propagate directly to the output through routing operations, whereas error propagation from the middle two branches involves second-order processes, and propagation from the rightmost branch constitutes a third-order process (see Supplementary Section II E for a detailed analysis). These observations indicate effective error localization within the bucket-brigade architecture. The noise resilience of the bucket-brigade architecture can be further understood from the perspective of quantum entanglement. Thus, we measure the von Neumann entanglement entropy of the router qubits after loading the address state $|++\rangle$ to the system. Experimentally, the entropy is obtained as $S = -\text{Tr}[\rho \log \rho]$, where ρ is the density matrix of the router qubit obtained through quantum state tomography. As depicted in Fig. 4b, the entanglement entropy decreases with increasing layer depth, reflecting the reduced quantum correlations between QRAM branches. This reduction in entanglement confines errors to their original branches, thus enhancing the overall noise resistance.

The inherent noise resilience of the bucket-brigade QRAM architecture makes it particularly promising for the scalable implementation of noisy quantum devices. To assess hardware requirements for scaling, we perform simulations evaluating the fidelity performance

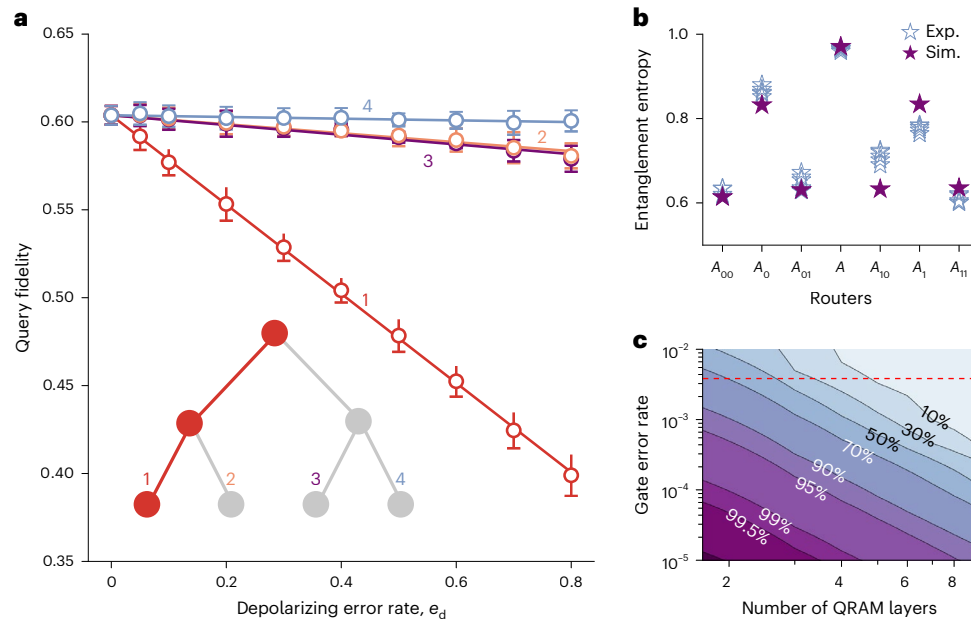


Fig. 4 | Error analysis of the three-layer bucket-brigade QRAM architecture. **a**, Experimental query fidelity under depolarizing errors with various error rates. The fidelity value is averaged over five repeated experiments, with the error bar representing the standard deviation. Each time, we inject the depolarizing errors into one of the router qubits at the third level after the data-loading operation (Extended Data Fig. 3). Inset: schematic of the QRAM with error injection sites labelled and the queried branch highlighted. **b**, Von Neumann entanglement entropy of different routers in the QRAM, measured after the address-loading

operation. The experimental results from five repeated runs are displayed along with the noisy simulation results. The x-axis labels for the router qubits correspond to the layout shown in Extended Data Fig. 3. **c**, The query fidelity as a function of the two-qubit gate error rate and number of layers in the bucket-brigade architecture, obtained from the noisy numerical simulation. The dashed line marks the gate error rate (3.8×10^{-3}) in the current work. Exp., experiment; Sim., simulation.

across architectures with two to nine layers while also scaling the gate error rates, as shown in Fig. 4c. To enable efficient numerical simulations (Methods), we consider only the depolarizing error channel and fix the error rate of the single-qubit gate to be one-sixth of that of the two-qubit gate. The simulations are conducted for a maximally superposed query address state $|++\dots+\rangle$ and a memory configuration where all classical data bits are set to 1. Figure 4c illustrates the contour of the query fidelity in the parameter space, which was derived by extrapolating the data in Extended Data Fig. 5. Notably, the required two-qubit gate error rate p to maintain a target query fidelity follows a power-law decay ($p = A \times L^\alpha$ with $\alpha \approx -2.728$) as the number of QRAM layers L increases, despite the exponential growth in system qubits (see Supplementary Section I A for further analysis). The results indicate that, with the future inclusion of quantum error correction⁴¹, the bucket-brigade QRAM will require a code distance that grows only logarithmically with respect to the number of layers L to maintain near-perfect fidelity.

Conclusion and outlook

In this work, we present the experimental realization of a scalable bucket-brigade QRAM architecture on a programmable superconducting quantum processor incorporating optimized routing operations and error mitigation protocols. By unveiling the localized nature of the error propagation mechanism and its physical origin, we provide experimental evidence of the inherent noise resilience in this architecture.

Furthermore, we numerically demonstrate that the bucket-brigade QRAM exhibits polylogarithmic scaling of the query infidelity with respect to gate errors, a key feature that supports the practical feasibility of implementing robust, large-scale QRAM on near-term quantum hardware, including devices with square lattice structures.

Looking ahead, as the size of QRAM grows, the exponential increase in the number of qubits required poses challenges to current superconducting processors, which necessitates the exploration of more hardware-efficient implementations. For example, it has

been proposed that hybrid quantum acoustic systems could be leveraged to realize QRAM in a hardware-efficient manner²⁴. Additionally, a distributed QRAM architecture would be desirable for alleviating the qubit and connectivity demands on individual quantum processors. In this context, we note that deterministic quantum state and gate teleportation between distant superconducting chips has recently been experimentally demonstrated⁴². Overall, analogous to random access memory in classical computers, it will be necessary to design a dedicated device for QRAM to tackle the exponential qubit overhead.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41567-026-03218-2>.

References

- Gupta, S. & Zia, R. Quantum neural networks. *J. Comput. Syst. Sci.* **63**, 355–383 (2001).
- Rebentrost, P., Mohseni, M. & Lloyd, S. Quantum support vector machine for big data classification. *Phys. Rev. Lett.* **113**, 130503 (2014).
- Bang, J., Dutta, A., Lee, S.-W. & Kim, J. Optimal usage of quantum random access memory in quantum machine learning. *Phys. Rev. A* **99**, 012326 (2019).
- Phalak, K., Li, J. & Ghosh, S. Trainable PQC-based QRAM for quantum storage. *IEEE Access* **11**, 51892 (2023).
- Liu, J. et al. Towards provably efficient quantum algorithms for large-scale machine-learning models. *Nat. Commun.* **15**, 434 (2024).
- Ciliberto, C. et al. Quantum machine learning: a classical perspective. *Proc. R. Soc. A* **474**, 20170551 (2018).

7. Berry, D. W., Gidney, C., Motta, M., McClean, J. R. & Babbush, R. Qubitization of arbitrary basis quantum chemistry leveraging sparsity and low rank factorization. *Quantum* **3**, 208 (2019).
8. Babbush, R. et al. Encoding electronic spectra in quantum circuits with linear T complexity. *Phys. Rev. X* **8**, 041015 (2018).
9. Childs, A. M., Leng, J., Li, T., Liu, J.-P. & Zhang, C. Quantum simulation of real-space dynamics. *Quantum* **6**, 860 (2022).
10. Mao, R., Tian, G. & Sun, X. Toward optimal circuit size for sparse quantum state preparation. *Phys. Rev. A* **110**, 032439 (2024).
11. Feniou, C. et al. Sparse quantum state preparation for strongly correlated systems. *J. Phys. Chem. Lett.* **15**, 3197 (2024).
12. Grover, L. K. A fast quantum mechanical algorithm for database search. In *Proc. Twenty-Eighth Annual ACM Symposium on Theory of Computing STOC '96* (ed Miller, G. L.) 212–219 (ACM, 1996).
13. Giovannetti, V., Lloyd, S. & Maccone, L. Quantum private queries. *Phys. Rev. Lett.* **100**, 230502 (2008).
14. Wiebe, N., Braun, D. & Lloyd, S. Quantum algorithm for data fitting. *Phys. Rev. Lett.* **109**, 050505 (2012).
15. Lloyd, S., Garnerone, S. & Zanardi, P. Quantum algorithms for topological and geometric analysis of data. *Nat. Commun.* **7**, 10138 (2016).
16. Harrow, A. W., Hassidim, A. & Lloyd, S. Quantum algorithm for linear systems of equations. *Phys. Rev. Lett.* **103**, 150502 (2009).
17. Aaronson, S. Read the fine print. *Nat. Phys.* **11**, 291 (2015).
18. Giovannetti, V., Lloyd, S. & Maccone, L. Quantum random access memory. *Phys. Rev. Lett.* **100**, 160501 (2008).
19. Giovannetti, V., Lloyd, S. & Maccone, L. Architectures for a quantum random access memory. *Phys. Rev. A* **78**, 052310 (2008).
20. Matteo, O. D., Gheorghiu, V. & Mosca, M. Fault-tolerant resource estimation of quantum random-access memories. *IEEE Trans. Quantum Eng.* **1**, 4500213 (2020).
21. Hann, C. T., Lee, G., Girvin, S. & Jiang, L. Resilience of quantum random access memory to generic noise. *PRX Quantum* **2**, 020311 (2021).
22. Paler, A., Oumarou, O. & Basmadjian, R. Parallelizing the queries in a bucket-brigade quantum random access memory. *Phys. Rev. A* **102**, 032608 (2020).
23. Xu, S., Hann, C. T., Foxman, B., Girvin, S. M. & Ding, Y. Systems architecture for quantum random access memory. In *Proc. MICRO '23* (ed Kobayashi, H.) 526–538 (ACM, 2023).
24. Hann, C. T. et al. Hardware-efficient quantum random access memory with hybrid quantum acoustic systems. *Phys. Rev. Lett.* **123**, 250501 (2019).
25. Chen, Z.-Y. et al. Efficient and error-resilient data access protocols for a limited-sized quantum random access memory. Preprint at <https://arxiv.org/abs/2303.05207> (2023).
26. Wang, Y., Alexeev, Y., Jiang, L., Chong, F. T. & Liu, J. Fundamental causal bounds of quantum random access memories. *npj Quantum Inf.* **10**, 71 (2024).
27. Arunachalam, S., Gheorghiu, V., Jochym-O'Connor, T., Mosca, M. & Srinivasan, P. V. On the robustness of bucket brigade quantum RAM. *New J. Phys.* **17**, 123010 (2015).
28. Bugalho, L. et al. Resource-efficient simulation of noisy quantum circuits and application to network-enabled QRAM optimization. *npj Quantum Inf.* **9**, 105 (2023).
29. Xu, S., Lu, A. & Ding, Y. Fat-tree QRAM: a high-bandwidth shared quantum random access memory for parallel queries. In *Proc. 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems*, Vol. 2 (eds Bulitko, V. & Storandt, S.) 390–406 (ACM, 2025).
30. Chen, K. C., Dai, W., Errando-Herranz, C., Lloyd, S. & Englund, D. Scalable and high-fidelity quantum random access memory in spin-photon networks. *PRX Quantum* **2**, 030319 (2021).
31. Wang, Z., Qiao, H., Cleland, A. N. & Jiang, L. Quantum random access memory with transmon-controlled phonon routing. *Phys. Rev. Lett.* **134**, 210601 (2024).
32. Weiss, D., Puri, S. & Girvin, S. Quantum random access memory architectures using 3D superconducting cavities. *PRX Quantum* **5**, 020312 (2024).
33. Miao, C. et al. Implementation of a quantum addressable router using superconducting qubits. *PRX Quantum* **6**, 040335 (2025).
34. Zhang, S. et al. Demonstrating coherent quantum routers for bucket-brigade quantum random access memory on a superconducting processor. Preprint at <https://arxiv.org/abs/2505.13958> (2025).
35. Hu, X.-M., Guo, Y., Liu, B.-H., Li, C.-F. & Guo, G.-C. Progress in quantum teleportation. *Nat. Rev. Phys.* **5**, 339 (2023).
36. Cruz, P. M. Q. & Murta, B. Shallow unitary decompositions of quantum Fredkin and Toffoli gates for connectivity-aware equivalent circuit averaging. *APL Quantum* **1**, 016105 (2024).
37. Botea, A., Kishimoto, A. & Marinescu, R. On the complexity of quantum circuit compilation. In *Proc. 11th Annual Symposium on Combinatorial Search*, Vol. 9 (eds Sampson, A., Kim, M. & Rossbach, C. J.) 138–142 (AAAI, 2018).
38. Arute, F. et al. Quantum supremacy using a programmable superconducting processor. *Nature* **574**, 505 (2019).
39. Boixo, S. et al. Characterizing quantum supremacy in near-term devices. *Nat. Phys.* **14**, 595 (2018).
40. Paler, A., Devitt, S., Nemoto, K. & Polian, I. Software-based Pauli tracking in fault-tolerant quantum circuits. In *2014 Design, Automation & Test in Europe Conference & Exhibition (DATE)* (eds Tahoori, M. & Ottavi, M.) <https://doi.org/10.7873/DATE.2014.137> (IEEE, 2014).
41. Acharya, R. et al. Quantum error correction below the surface code threshold. *Nature* **638**, 920 (2025).
42. Qiu, J. et al. Deterministic quantum state and gate teleportation between distant superconducting chips. *Sci. Bull.* **70**, 351 (2025).

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Methods

Numerical simulation

The error budget analyses of the two- and three-layer QRAM were performed through numerical simulations under various physical error models using the Qiskit package⁴³. However, simulating QRAM with more layers becomes challenging due to the exponential growth in classical memory requirements. To address this and enable scaling analyses, we used an efficient classical algorithm for the QRAM simulation. For a QRAM memory of size N , conventional simulation methods based on the full state vector (such as those used in our error budget analyses) require a complexity of $\mathcal{O}(2^N)$. By contrast, our approach requires only a complexity of $\mathcal{O}(N_A \log N)$, where $N_A (\leq N)$ denotes the number of queried branches in QRAM. The efficiency arises because the QRAM circuits are examples of efficiently computable sparse operations^{21,44}. Specifically, the initial state of the whole system, including the router qubits, can be expressed as a superposition of N_A computational basis states:

$$\sum_{i=0}^{N_A-1} \alpha_i |\varphi_i\rangle_A |0\rangle_D |0\rangle_R,$$

where subscripts A, D and R denote the subsystems composed of address, data and router qubits, respectively, and φ_i enumerates the non-zero components in the quantum address. The QRAM circuit is composed of gates from the gate set $\mathcal{G} = \{\text{SWAP}, U_{\text{CSWAP}}, H, X\}$, where U_{CSWAP} denotes the functionally equivalent CSWAP gate implemented in this work. These gates preserve the state of each qubit within the set $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$ for every computational basis state. This crucial property enables efficient tracking of the state evolution of the system for each address component during the entire QRAM operation.

We simulated the noisy QRAM circuit using the Monte Carlo method. We considered Pauli error channels for each gate in $\mathcal{G}$, which keep the efficiently computable sparse properties of the original circuit and enable an efficient noisy simulation. For single-qubit gates H and X, we employed the depolarizing error model, where an erroneous X, Y or Z gate was applied to the qubit after the gate, each with a probability of p_s (also referred to as the Pauli error rate). To simulate the noise of the SWAP and U_{CSWAP} gates, we applied erroneous Pauli gates to the corresponding qubits after the gate, each of which was sampled with a probability calculated based on employing the depolarizing error model with Pauli error rates of p_s and p for the single- and two-qubit gates in the corresponding decomposition circuits (see Supplementary Information for details). In our simulation, we set $p_s = p/6$. The gate error rate in Fig. 4c corresponds to the value of p .

Quantum teleportation

The embedding of an L -layer QRAM in a two-dimensional square lattice requires further $\mathcal{O}(2^L)$ ancillary qubits to be filled in between layers of the QRAM. As a result, the SWAP-based communication scheme between adjacent layers can lead to a large query latency that increases the time complexity of the entire QRAM circuit. Quantum teleportation offers an elegant way to escape this dilemma. In our experiment, we present a proof-of-principle demonstration of the teleportation-based communication scheme in the QRAM circuit. We used quantum teleportation in the data-retrieval phase of the two-layer QRAM, where the quantum information was transferred back from the data router qubit (labelled D in Fig. 1d) to the data register qubit (labelled as d) through an ancilla. Specifically, upon routing the data information into qubit D , we prepared the ancilla and qubit d in the Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ and then performed the Bell measurement on D and the ancilla. Based on the measurement output of D and the ancilla, we tracked the required Pauli operations on d and then virtually applied them via a reinterpretation of the measurement outcomes in the state tomography of d . For example, when both D and the ancilla are measured in the $|1\rangle$ state, the operator to be applied to d is iY . This operator preserves the measurement outcomes in the Y basis while flipping those in the X and Z bases.

Accordingly, we reinterpret the tomography measurement in the X (or Z) basis as having been performed in the $-X$ (or $-Z$) basis. Note that the Pauli-tracking procedure could be replaced with a classical feed-forward operation to guarantee the scalability of the algorithm.

Data availability

The data presented in the figures and that support the other findings of this study are publicly available via Zenodo at <https://doi.org/10.5281/zenodo.18408759> (ref. 45.) Source data are provided with this paper.

Code availability

The data analysis and numerical simulation codes for this study are publicly available via Zenodo at <https://doi.org/10.5281/zenodo.18408759> (ref. 45.).

References

43. Javadi-Abhari, A. et al. Quantum computing with Qiskit. Preprint at <https://arxiv.org/abs/2405.08810> (2024).
44. Van den Nest, M. Simulating quantum computers with probabilistic methods. *Quant. Inf. Comp.* **11**, 784–812 (2011).
45. Shen, F., Ji, Y. & Xiang, D. Source data and simulation code for ‘a bucket-bridge quantum random access memory’. *Zenodo* <https://doi.org/10.5281/zenodo.18408759> (2026).

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Author contributions

C.S., L.L., Z.W. and J. Yin conceived the project. F.S. conducted the experiments under the supervision of C.S. Y.J. designed the experimental circuits and performed the theoretical analyses under the supervision of C.S. Y.J., F.S. and D.X. performed the simulations and analysed the experimental data under the supervision of C.S. and L.L. H.L. fabricated the device. C.S., Z.W., Q.G., H.L., P.Z., Y.W., K.W., C.Z., A.Z., Y.Z., Y.G., Z.C., G.L., J. Yang and Y.H. contributed to the experimental set-up. Y.J., F.S., D.X., L.L. and C.S. wrote the paper with input from all authors. All authors discussed the results and contributed to the final version of the paper.

Competing interests

The authors declare no competing interests.

Additional information

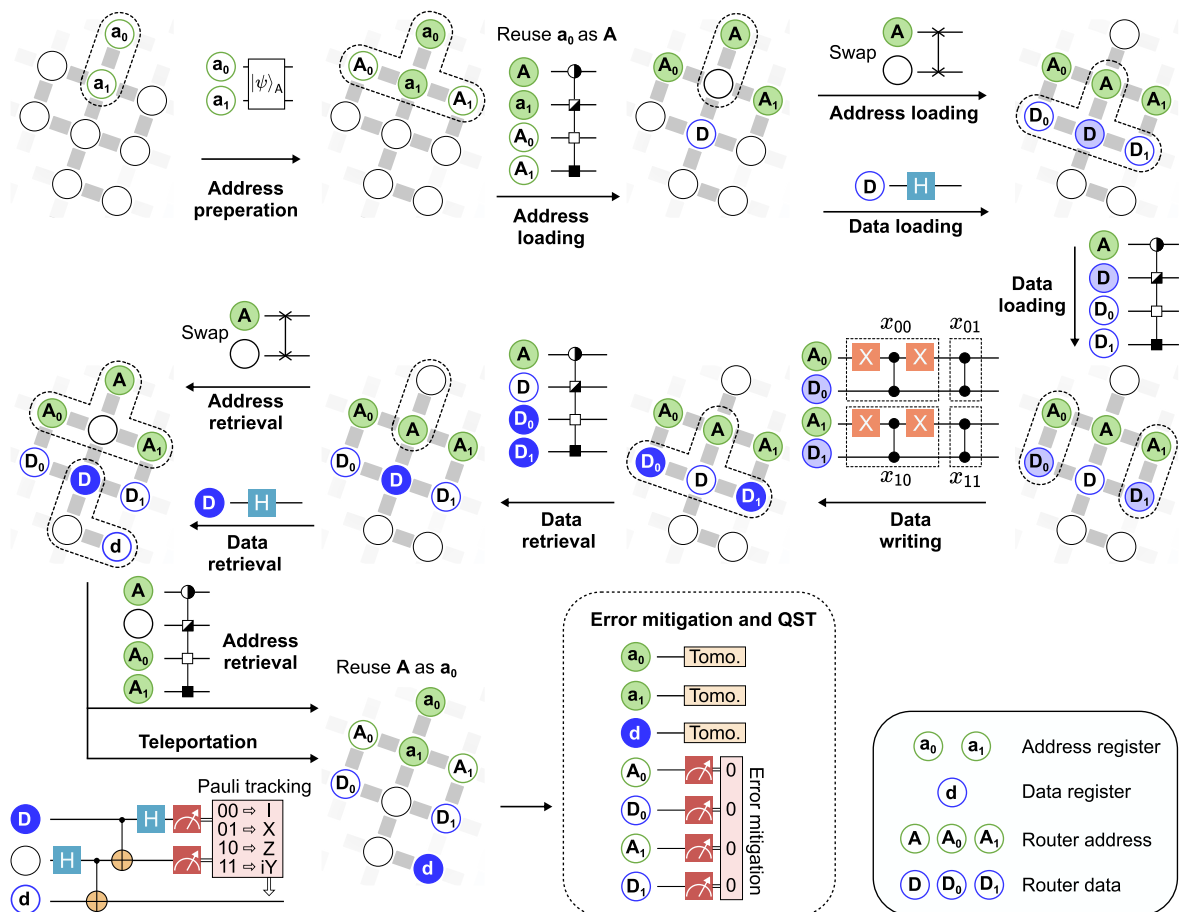
Extended data is available for this paper at <https://doi.org/10.1038/s41567-026-03218-2>.

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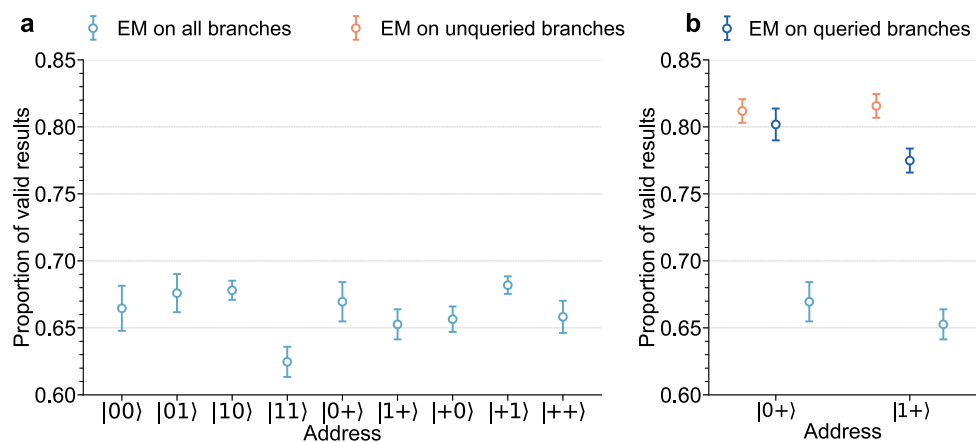
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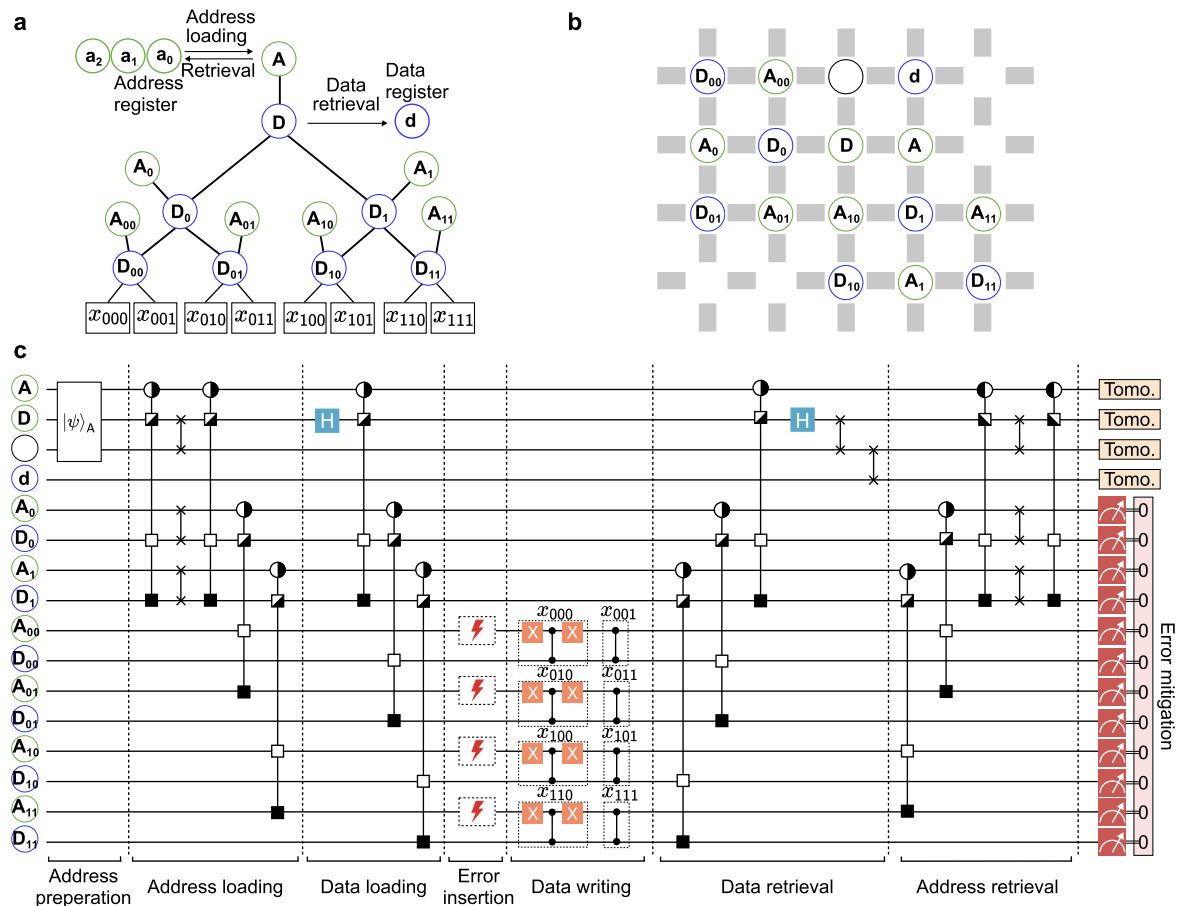
Extended Data Fig. 1 | Illustration of the query process of two-layer QRAM. which provides a detailed explanation of Fig. 1d, elucidating the role of each physical qubit in every phase and the operations performed. **Address preparation:** The initial address state is prepared on the two address register qubits, a_0 and a_1 . **Address loading:** The qubit originally holding a_0 is reassigned as the router address qubit A . A quantum routing operation is then applied to load the information from a_1 into the second-layer router address qubits A_0 and A_1 . Finally, a SWAP gate moves A to the qubit originally occupied by a_1 . **Data loading:** The router data qubit D is initialized to $|+\rangle$ to enable phase encoding during data writing. Its state is then loaded into the second-layer router data qubits D_0 and D_1 via a quantum routing operation. **Data writing:** Classical bits

are encoded by selectively applying or omitting the quantum gates inside the dashed squares, depending on the bit value (1 or 0). **Data retrieval:** The quantum state is retrieved from D_0 and D_1 back to D via quantum routing. A Hadamard gate is applied to D to return the state to the computational basis. Quantum teleportation is then used to transfer the state from D to the data register qubit d . **Address retrieval:** A is swapped back to the qubit originally holding a_0 . The address information in A_0 and A_1 is then retrieved back to the qubit initially holding a_1 via a quantum routing operation. **Error mitigation and quantum state tomography (QST):** QST is applied to a_0, a_1 , and d , with the tomography operation on d reinterpreted via Pauli tracking (see Methods). Router qubits are also measured for post-selection error mitigation.



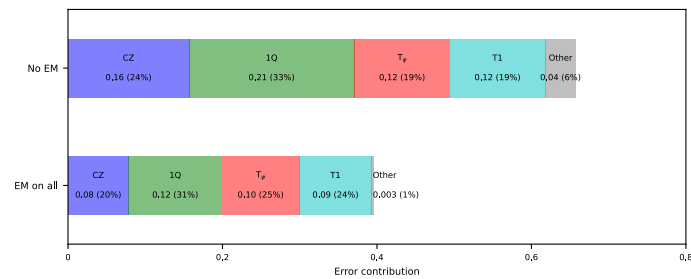
Extended Data Fig. 2 | The proportion of valid results after error mitigation in the two-layer bucket-brigade QRAM. In all figures, the proportion values represent the average value of the proportion of valid results across the 16 classical data configurations, with the error bar representing their standard

deviation. **a**, The proportion of valid results after error mitigation for a two-layer bucket brigade QRAM under varying query addresses. **b**, The proportion of valid results for different error mitigation strategies under query addresses $|0+\rangle$ and $|1+\rangle$.



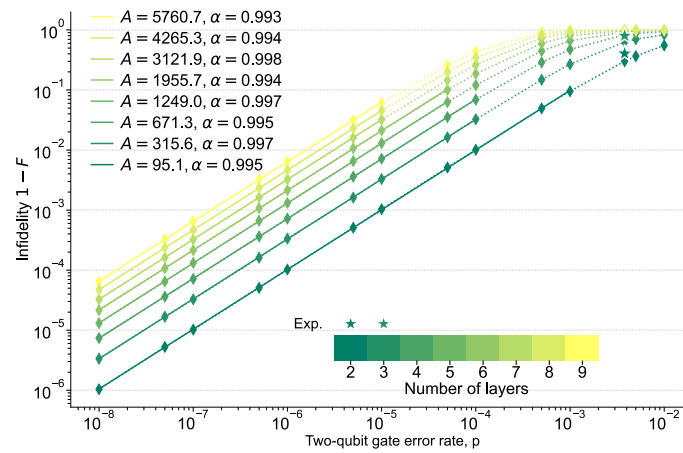
Extended Data Fig. 3 | Experimental construction of the three-layer bucket-brigade QRAM. **a**, Schematic of the three-layer bucket-brigade QRAM, which can query eight classical data bits, labeled from x_{000} to x_{111} , according to the query address. **b**, Device layout for implementing the three-layer bucket-brigade QRAM. Physical qubits are labeled according to their assigned roles during operational phases from address loading to address retrieval. **c**, Quantum circuit for the error scaling experiment of the three-layer QRAM. As in the two-layer QRAM case, the router address qubit (A), router data qubit (D), and one ancilla qubit are reused as the address register a_0, a_1 , and a_2 at the beginning and end of the experiment. Instead of using quantum teleportation, here the data bus is directly transferred from the router data qubit D to the data register qubit d via two SWAP gates at the end of the data retrieval phase. **a**, Schematic of the

three-layer bucket-brigade QRAM, which can query eight classical data bits, labeled from x_{000} to x_{111} , according to the query address. **b**, Device layout for implementing the three-layer bucket-brigade QRAM. Physical qubits are labeled according to their assigned roles during operational phases from address loading to address retrieval. **c**, Quantum circuit for the error scaling experiment of the three-layer QRAM. As in the two-layer QRAM case, the router address qubit (A), router data qubit (D), and one ancilla qubit are reused as the address register a_0, a_1 , and a_2 at the beginning and end of the experiment. Instead of using quantum teleportation, here the data bus is directly transferred from the router data qubit D to the data register qubit d via two SWAP gates at the end of the data retrieval phase.



Extended Data Fig. 4 | Error budget for the three-layer QRAM query infidelity. The results are obtained via numerical simulations with address state $|00+\rangle$ and the classic configuration 01010101. The percentage in parentheses represents the proportion of this error component in the total error. CZ: CZ gate

depolarizing error, excluding the decoherence error during CZ gates; 1Q: Single-qubit gate depolarizing error, excluding the decoherence error during single-qubit gates; T1: Qubit energy relaxation error; T_φ: Qubit dephasing error. See Supplementary Information for details on the error budget analysis.



Extended Data Fig. 5 | Simulated query infidelity versus two-qubit gate error rate. The simulation data are fitted to $1 - F = A \times p^\alpha$, where F is the query fidelity and p is the two-qubit gate error rate. Only data points with infidelity below 0.1 are included in the fit. Experimental results for both two- and three-layer

QRAM at $p = 3.8 \times 10^{-3}$ are marked by pentagrams. The discrepancy between the experimental and numerical results is mainly due to the over-simplification of the error models employed in the numerical simulation (see Method).